Zhaoyi Wang

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SUMMARY

AI-driven remote sensing and 3D spatial analysis for environmental and ecosystem monitoring, with a focus on LiDAR point clouds, photogrammetry, and multi-temporal change detection applied to natural systems, and with methods transferable to terrain dynamics, vegetation-related analysis, and forest-relevant environments.

EDUCATION

ETH Zurich — **Zurich, Switzerland**
PhD in Geomatics, Institute of Geodesy and Photogrammetry — Oct. 2021 - Nov. 2025
- Thesis: "Cross-modal fusion of point clouds and RGB images for landslide monitoring"

University of Science and Technology of China — **Hefei, China**
MSc in Precision Engineering Geodesy — Sep. 2018 - Jun. 2021

Changsha University of Science and Technology — **Changsha, China**
BSc in Surveying and Mapping Engineering — Sep. 2014 - Jun. 2018

PROFESSIONAL EXPERIENCE

Geosensors and Engineering Geodesy, ETH Zurich — **Zurich, Switzerland**
Research Assistant — Oct. 2021 - Dec. 2025
- Developed AI-enabled geospatial pipelines for environmental monitoring, projecting TLS point clouds into panoramic images and detecting artificial targets using customized deep learning detectors (Faster R-CNN, YOLO-X, YOLO-v7)
- Proposed cross-modal feature fusion methods and developed end-to-end geospatial pipelines for robust point cloud registration and multi-epoch 3D displacement estimation in complex natural environments, integrating TLS point clouds and RGB images
- Implemented GPU-accelerated algorithms (PyTorch + CUDA) for large-scale environmental 3D data processing
- Designed automated workflows for preprocessing, depth map consistency, visibility tracking, and evaluation across multi-epoch UAV datasets over natural terrain and environmental monitoring scenarios
- Application focus: environmental monitoring, natural terrain and vegetation dynamics, and ecosystem-relevant 3D change analysis

3DOM, Fondazione Bruno Kessler (FBK) — **Trento, Italy**
Visiting Researcher — Sep. 2025 - Nov. 2025
- Developed and benchmarked multi-scene, multi-sensor 3D change detection methods, focusing on generalization across diverse spatial scales, point densities, and environmental conditions (e.g., terrain complexity, vegetation cover)
- Contributed to automated pipelines for multi-epoch UAV RGB reconstruction and 3D surface change detection for UAV-based landslide monitoring

Dilusense Co., Ltd. — **Hefei, China**
Computer Vision Algorithm Intern — Mar. 2021 - Jun. 2021
- Developed algorithms for RGB-D camera calibration and depth estimation in structured light systems

ADDITIONAL EXPERIENCE

- Teaching Assistant for Geomonitoring and Geosensors (2022), Geospatial Data Acquisition (2023, 2024)
- Volunteer Interpreter, University of Science and Technology of China (Sep. 2018 - Jun. 2021)
- International Geoinformatics Summer School, Wuhan University (Aug. 2020)
- Exchange Program, Hong Kong Polytechnic University (Jul. 2017)

SELECTED ACTIVITIES AND ACHIEVEMENTS

- Best Presentation Award, ISPRS Geospatial Week – Laser Scanning Worknshop (2023)
- National Scholarship, University of Science and Technology of China, Top 2% (2020)
- Top Grade Scholarship, Changsha University of Science and Technology, Top 1% (4×), 2015–2018
- Reviewer: ISPRS Journal of Photogrammetry and Remote Sensing, *International Journal of Applied Earth Observation and Geoinformation*, *Measurement*, *Computational Urban Science*, *and The European Journal on Artificial Intelligence*

CORE SKILLS

- Application Domains: Environmental monitoring, ecosystem and vegetation analysis, geohazard and terrain dynamics
- Remote Sensing & 3D Spatial Data: LiDAR (TLS/UAV), Photogrammetry, SfM/MVS, multi-temporal 3D change detection, geohazard monitoring, object detection, geometric feature extraction
- AI & Programming: Python, C++, Shell scripting, MATLAB, CUDA, Git, LaTeX, Linux, PyTorch, Open3D, OpenCV, COLMAP
- Software: Leica Cyclone, RIEGL RiSCAN, FARO SCENE, Agisoft Metashape, CloudCompare, Unity
- Languages: Mandarin (Native), English (Professional)
- Extracurricular: Half-marathon running, skiing, hiking, table tennis

SELECTED PUBLICATIONS AND SUBMISSIONS

Wang, Z.Y., Trybala, P., Wieser, A., & Remondino, F. (2026). MultiChange3D: A multi-scene, multi-sensor dataset for benchmarking 3D geometric change detection. Under review.

Wang, Z.Y., Butt, J.A., Huang, S.Y., Medić, T., & Wieser, A. (2026). Dense 3D displacement estimation for landslide monitoring via fusion of TLS point clouds and embedded RGB images. *International Journal of Applied Earth Observation and Geoinformation*.

Wang, Z.Y., Butt, J.A., Huang, S.Y., Meyer, N., Medić, T., & Wieser, A. (2025). An approach for RGB-guided dense 3D displacement estimation in TLS-based geomonitoring. *ISPRS Annals, ISPRS Geospatial Week 2025*. (Oral presentation)

Wang, Z.Y., Huang, S.Y., Butt, J.A., Cai, Y.Z., Varga, M., & Wieser, A. (2025). Cross-modal feature fusion for robust point cloud registration with ambiguous geometry. *ISPRS Journal of Photogrammetry and Remote Sensing.*

Wang, Z.Y., Varga, M., Medić, T., & Wieser, A. (2023). Assessing the alignment between geometry and colors in TLS colored point clouds. *ISPRS Annals, ISPRS Geospatial Week 2023*. (Oral presentation)

References available upon request: Prof. Dr. Andreas Wieser, Prof. Dr. Konrad Schindler, Prof. Dr. Fabio Remondino